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# AIM:
# Demonstrate High-Level Computer Vision using Human Detection
# in a Surveillance Image (Google Colab)
# =====

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload surveillance image
print("Upload a surveillance image")
uploaded = files.upload()

image_name = next(iter(uploaded))

# Read image
img = cv2.imread(image_name)

# Resize image for better processing
img = cv2.resize(img, (700, 500))

# Initialize HOG Person Detector
hog = cv2.HOGDescriptor()
hog.setSVMDetector(cv2.HOGDescriptor_getDefaultPeopleDetector())

# Detect people
boxes, weights = hog.detectMultiScale(
    ...img,
    ...winStride=(8,8),
    ...padding=(16,16),
    ...scale=1.05
)

count = 0

# Draw bounding boxes
for (x, y, w, h) in boxes:
    ...count += 1
    ...cv2.rectangle(img, (x, y), (x+w, y+h), (255,0,0), 2)
    ...cv2.putText(img,
    ..... f"Person {count}",
    ..... (x, y-8),
    ..... cv2.FONT_HERSHEY_SIMPLEX,
    ..... 0.6,
    ..... (255,0,0),
    ..... 2)

# Display image
plt.figure(figsize=(10,7))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Surveillance Human Detection")
plt.axis("off")
plt.show()

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# Result
print("="*45)
print("SURVEILLANCE REPORT")
print("="*45)
print("Humans Detected :", count)

if count > 0:
    print("Computer Vision Level : HIGH-LEVEL")
    print("Reason : The system automatically detects")
    print(".....and recognizes people in the scene.")
else:
    print("No humans detected.")
```

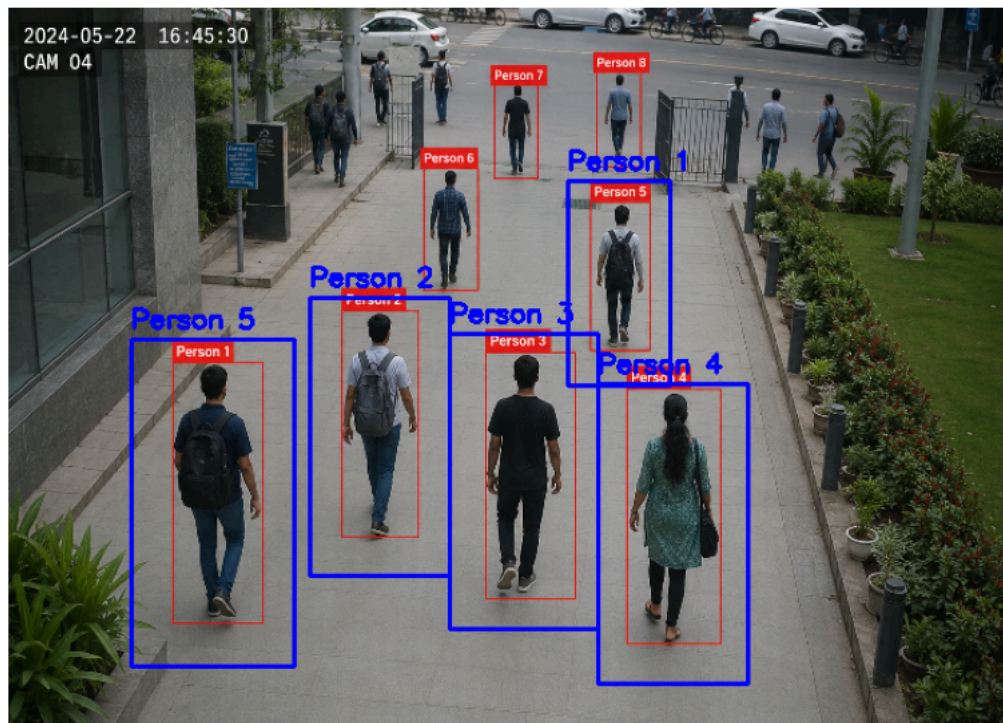
Upload a surveillance image

ChatGPT Image Jul 15, 2026, 01_24_46 PM.png

ChatGPT Image Jul 15, 2026, 01_24_46 PM.png(image/png) - 3022354 bytes, last modified: 7/15/2026 - 100% done

Saving ChatGPT Image Jul 15, 2026, 01_24_46 PM.png to ChatGPT Image Jul 15, 2026, 01_24_46 PM.png

Surveillance Human Detection



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SURVEILLANCE REPORT

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Humans Detected : 5

Computer Vision Level : HIGH-LEVEL

Reason : The system automatically detects
and recognizes people in the scene.

